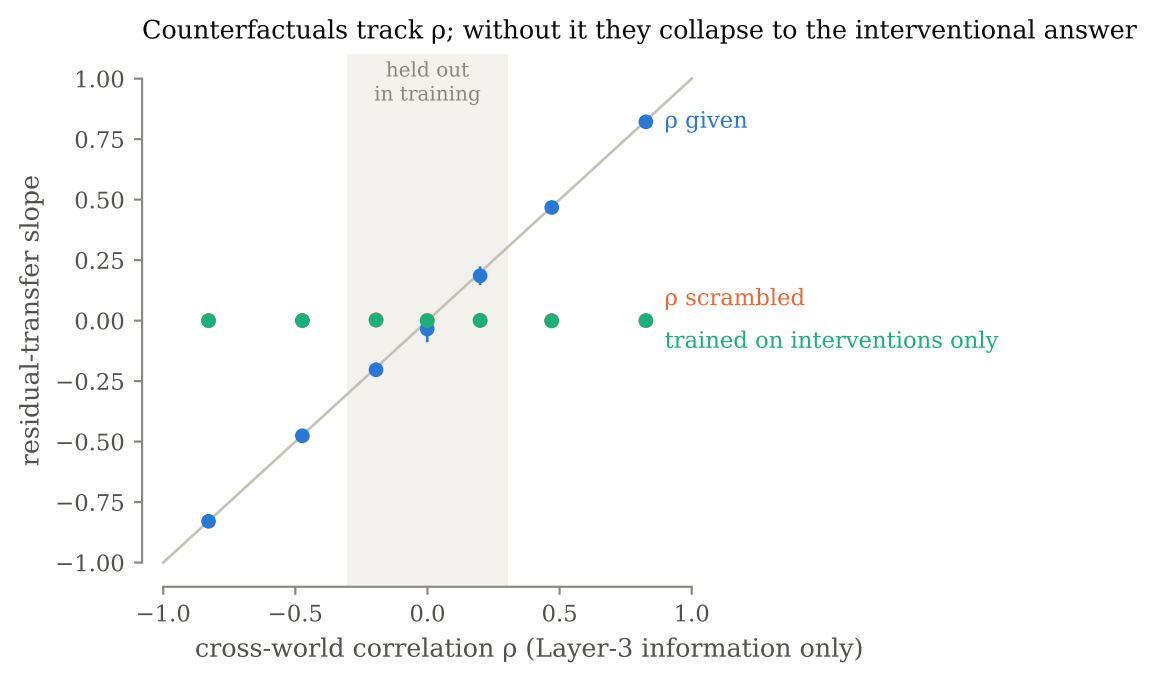
Counterfactuals track ρ; without it they collapse to the interventional answer
held out
in training
ρ given
ρ scrambled
trained on interventions only
residual-transfer slope
cross-world correlation ρ (Layer-3 information only)
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
−1.0
−0.5
0.0
0.5
1.0